THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET A

Chapter 4: **Exploring Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $9x^2 + 24x + 16$	$(a+b)^2 = a^2 + 2ab + b^2$
2	**(d)** $49x^2 + 14x + 1$	$(a+b)^2 = a^2 + 2ab + b^2$
3	**(b)** 1849	$43^2 = (40+3)^2$
4	**(b)** $(3x+4)^2$	identify $a^2 + 2ab + b^2 = (a+b)^2$
5	**(d)** $x^2 + 7x + 12$	$(x+a)(x+b) = x^2 + (a+b)x + ab$
6	**(b)** $(x+5)(x+6)$	find a, b with $a+b$ = middle coefficient and ab = constant
7	**(c)** $(x-7)(x-6)$	find a, b with $a+b$ = middle coefficient and ab = constant
8	**(d)** $x^3 + 6x^2 + 12x + 8$	$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
9	**(a)** $3xyz$	from $x^3 + y^3 + z^3 - 3xyz = (x+y+z)(\dots)$
10	**(d)** 3025	$60 \times 50 + 25$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$49x^2 + 56xy + 16y^2$	2
12	$\frac{25p^2}{4} + \frac{15pq}{2} + \frac{9q^2}{4}$	2
13	$9s^2 - 24st + 16t^2$	2
14	$x^2 + xy + \frac{y^2}{4}$	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Write as $(3x)^2 + (2y)^2 + z^2 + 2(3x)(2y) + 2(3x)(z) + 2(2y)(z) = (3x+2y+z)^2$. Answer: $(3x+2y+z)^2$	3
16	$p^2 + 6pq + 14pr + 9q^2 + 42qr + 49r^2$	3
17	$9x^2 - 12xy + 24xz + 4y^2 - 16yz + 16z^2$	3
	OR: $a = 2x,\ b = 3$: $8x^3 + 36x^2 + 54x + 27$	3
18	$27m^3 - 54m^2n + 36mn^2 - 8n^3$	3
	OR: $(2n)^3 - 3(2n)^2(5m) + 3(2n)(5m)^2 - (5m)^3 = (2n-5m)^3$.	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $25x^2 + 20xy + 4y^2$; $43^2 = (40+3)^2 = 1849$. (ii) $(a+b)^2 - (a^2+b^2) = 2ab > 0$ when $ab > 0$. For $ab < 0$, e.g. $a = 3, b = -1$: $4 < 10$.	2+3
	OR: (i) $25 + 49 - 72 = 2$. (ii) $(n-1)^2 + (n+1)^2 - 2n^2 = 2$ (2). (iii) $(n-2)^2 + (n+2)^2 - 2n^2 = 8$ — a constant 8, not 2.	1+2+2
20	(i) $(3a - b + 2c)^2$ [check: $9a^2 + b^2 + 4c^2 - 6ab + 12ac - 4bc$ ✓]. (ii) $(4s - 5t)^2$.	3+2
	OR: (i) $x^2 + 8x + 15 = (x+3)(x+5)$. (ii) $x(x-4) = 96 \Rightarrow (x-12)(x+8) = 0$, so $x = 12$ m, breadth 8 m. (iii) $2(12+8) = 40$ m.	1+3+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME — SET B

Chapter 4: **Exploring Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $4x^2 + 20x + 25$	$(a+b)^2 = a^2 + 2ab + b^2$
2	**(c)** $4x^2 - 12x + 9$	$(a-b)^2 = a^2 - 2ab + b^2$
3	**(c)** 841	$29^2 = (30-1)^2$
4	**(d)** $(2s+5t)^2$	identify $a^2 + 2ab + b^2 = (a+b)^2$
5	**(a)** $x^2 + 3x - 10$	$(x+a)(x+b) = x^2 + (a+b)x + ab$
6	**(d)** $(x-3)(x-2)$	find a, b with $a+b$ = middle coefficient and ab = constant
7	**(a)** $49x^2 - 4$	$(a+b)(a-b) = a^2 - b^2$
8	**(d)** $8x^3 - 12x^2 + 6x - 1$	$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$
9	**(d)** $x^3 - y^3$	standard identity
10	**(d)** 2	$(n-1)^2 + (n+1)^2 - 2n^2 = 2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(3x+4y)^2$	2
12	$(2s+5t)^2$	2
13	$(4y-3)^2$	2
14	$(7x+y)^2$	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$(p)^3 + 3p^2(2q) + 3p(2q)^2 + (2q)^3 = (p+2q)^3$, so the side is $p+2q$.	3
16	Numerator $= (x-3)(x-4)$; denominator $= 5(x^2+x-20) = 5(x-4)(x+5)$. Cancelling $(x-4)$: $\dfrac{x-3}{5(x+5)}$.	1+1+1
17	Numerator $= (6s-t)^2$. Denominator: $a+b = 2s,\ ab = -48s^2 \Rightarrow (t+8s)(t-6s) = -(6s-t)(t+8s)$. So the fraction $= \dfrac{(6s-t)^2}{(t-6s)(t+8s)} = -\dfrac{6s-t}{t+8s} = \dfrac{t-6s}{t+8s}$.	1+1+1
	OR: $p^4 - 16 = (p^2-4)(p^2+4) = (p-2)(p+2)(p^2+4)$; denominator $= (p-2)^2$. Result: $\dfrac{(p+2)(p^2+4)}{p-2}$.	1+1+1
18	If $x+y+z = 0$ the right side is 0 ⇒ $x^3 + y^3 + z^3 = 3xyz$. Here $-5+2+3 = 0$, so the sum $= 3(-5)(2)(3) = -90$. (Check: $-125+8+27 = -90$ ✓)	1+1+1
	OR: $(x+y+z)^2 = 100 = 38 + 2(xy+yz+zx) \Rightarrow xy+yz+zx = 31$.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $x^3 + 6x^2 + 12x + 8$. (ii) $(x-2)(x^2+2x+4)$. (iii) $10^6 + 3\times10^4\times3 + 3\times100\times9 + 27 = 1000000 + 90000 + 2700 + 27 = 1092727$.	1+2+2
	OR: (i) $100 = 38 + 2(xy+yz+zx) \Rightarrow 31$. (ii) $x^3+y^3+z^3 - 3xyz = (10)(38-31) = 70 \Rightarrow x^3+y^3+z^3 = 70 + 75 = 145$.	2+3
20	(i) Numerator $= 3(p-3q)(p+2q)$; denominator $= (p+5q)(p-2q)$ — check: $p^2 + 3pq - 10q^2 = (p+5q)(p-2q)$. Result $\dfrac{3(p-3q)(p+2q)}{(p+5q)(p-2q)}$. (ii) $\dfrac{(2y-5z)^2}{(5z-2y)(5z+2y)} = \dfrac{5z-2y}{5z+2y}$.	3+2
	OR: (i) $35^2 = 40\times30 + 25 = 1225$; $65^2 = 70\times60 + 25 = 4225$; $85^2 = 90\times80 + 25 = 7225$. (ii) Each ends in 25; the leading digits are $n(n+1)$ for numbers $n5$ ($3\times4 = 12$, $6\times7 = 42$, $8\times9 = 72$). (iii) $105^2 = 110\times100 + 25 = 11025$.	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.